

Sequences & Series

ANSWER KEY



Arithmetic sequences

- 1 For the arithmetic sequence with $a_1 = 9$ and common difference $d = 6$, find a_{10} .

$$a_{10} = a_1 + (10 - 1)d = 9 + 9(6) = 63.$$

- 2 For the arithmetic sequence with $a_1 = 15$ and $d = -4$, find a_{11} .

$$a_{11} = 15 + 10(-4) = 15 - 40 = -25.$$

Arithmetic series

- 3 Find S_{10} for the arithmetic sequence with $a_1 = 9$, $d = 6$ (using $a_{10} = 63$ from above).

$$S_{10} = \frac{n(a_1 + a_n)}{2} = \frac{10(9 + 63)}{2} = 5(72) = 360.$$

Geometric sequences

- 4 For the geometric sequence with $a_1 = 6$ and common ratio $r = 2$, find a_8 .

$$a_8 = a_1 r^7 = 6(2^7) = 6(128) = 768.$$

- 5 A geometric sequence has $a_1 = 2$ and common ratio $r = 5$. Find the smallest n for which $a_n = 1250$.

$$a_n = 2 \cdot 5^{n-1}. \text{ Setting } 2 \cdot 5^{n-1} = 1250 \text{ gives } 5^{n-1} = 625 = 5^4, \text{ so } n - 1 = 4 \text{ and } n = 5.$$

Geometric series (finite)

- 6 Find S_8 for the geometric sequence with $a_1 = 6$, $r = 2$ (using $a_8 = 768$ from above).

$$S_8 = \frac{a_1(r^8 - 1)}{r - 1} = \frac{6(256 - 1)}{2 - 1} = 6(255) = 1530.$$

Infinite geometric series

- 7 Find the sum of the infinite geometric series $36 + 12 + 4 + \dots$.

$$a = 36, r = \frac{1}{3}, \text{ and } |r| < 1, \text{ so the sum is } \frac{36}{1 - \frac{1}{3}} = \frac{36}{\frac{2}{3}} = 54.$$

- 8 Find the sum of the infinite geometric series $12 - 4 + \frac{4}{3} - \dots$.
 $a = 12$, $r = -\frac{1}{3}$, and $|r| < 1$, so the sum is $\frac{12}{1 - (-\frac{1}{3})} = \frac{12}{\frac{4}{3}} = 9$.
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Multiple choice — easy

- 9 For the arithmetic sequence with $a_1 = 0$ and common difference $d = 9$, find a_8 .

a. 63 ✓

b. 65

c. 54

d. 72

$$a_8 = a_1 + (8 - 1)d = 0 + 7(9) = 63.$$

- 10 For the arithmetic sequence with $a_1 = 17$ and common difference $d = -3$, find a_9 .

a. -7 ✓

b. -4

c. -5

d. -10

$$a_9 = a_1 + (9 - 1)d = 17 + 8(-3) = -7.$$

- 11 For the arithmetic sequence with $a_1 = -11$ and common difference $d = -5$, find a_6 .

a. -36 ✓

b. -34

c. -41

d. -31

$$a_6 = a_1 + (6 - 1)d = -11 + 5(-5) = -36.$$

12 For the arithmetic sequence with $a_1 = -4$ and common difference $d = -5$, find a_7 .

a. -34 ✓

b. -29

c. -39

d. -32

$$a_7 = a_1 + (7 - 1)d = -4 + 6(-5) = -34.$$

13 For the arithmetic sequence with $a_1 = -6$ and common difference $d = 6$, find a_7 .

a. 36

b. 30 ✓

c. 32

d. 24

$$a_7 = a_1 + (7 - 1)d = -6 + 6(6) = 30.$$

14 For the arithmetic sequence with $a_1 = -17$ and common difference $d = -6$, find a_{11} .

a. -83

b. -77 ✓

c. -71

d. -75

$$a_{11} = a_1 + (11 - 1)d = -17 + 10(-6) = -77.$$

15 For the arithmetic sequence with $a_1 = 15$ and common difference $d = -9$, find a_7 .

a. -30

b. -48

c. -37

d. -39 ✓

$$a_7 = a_1 + (7 - 1)d = 15 + 6(-9) = -39.$$

16 For the arithmetic sequence with $a_1 = 15$ and common difference $d = -2$, find a_{12} .

a. -7 ✓

b. -9

c. -4

d. -5

$$a_{12} = a_1 + (12 - 1)d = 15 + 11(-2) = -7.$$

17 For the arithmetic sequence with $a_1 = 1$ and common difference $d = -6$, find a_{10} .

a. -51

b. -47

c. -53 ✓

d. -59

$$a_{10} = a_1 + (10 - 1)d = 1 + 9(-6) = -53.$$

18 For the arithmetic sequence with $a_1 = -6$ and common difference $d = -1$, find a_7 .

a. -13

b. -11

c. -10

d. -12 ✓

$$a_7 = a_1 + (7 - 1)d = -6 + 6(-1) = -12.$$

19 For the arithmetic sequence with $a_1 = -15$ and common difference $d = 2$, find a_6 .

- a. -3
- b. -5 ✓
- c. -7
- d. -2

$$a_6 = a_1 + (6 - 1)d = -15 + 5(2) = -5.$$

20 For the arithmetic sequence with $a_1 = -3$ and common difference $d = 9$, find a_7 .

- a. 53
- b. 42
- c. 51 ✓
- d. 60

$$a_7 = a_1 + (7 - 1)d = -3 + 6(9) = 51.$$

21 For the arithmetic sequence with $a_1 = 12$ and common difference $d = -6$, find a_6 .

- a. -12
- b. -18 ✓
- c. -24
- d. -16

$$a_6 = a_1 + (6 - 1)d = 12 + 5(-6) = -18.$$

22 For the arithmetic sequence with $a_1 = 4$ and common difference $d = -3$, find a_{13} .

- a. -30
- b. -32 ✓
- c. -35
- d. -29

$$a_{13} = a_1 + (13 - 1)d = 4 + 12(-3) = -32.$$

23 For the arithmetic sequence with $a_1 = -16$ and common difference $d = -9$, find a_7 .

- a. -68
- b. -61
- c. -70 ✓
- d. -79

$$a_7 = a_1 + (7 - 1)d = -16 + 6(-9) = -70.$$

Multiple choice — medium

24 Find S_6 for the arithmetic sequence with $a_1 = -13$, $d = -1$.

- a. -94
- b. -89
- c. -97
- d. -93 ✓

$$a_6 = -13 + (5)(-1) = -18. S_6 = \frac{6(-13 - 18)}{2} = 3(-31) = -93.$$

25 Find S_{12} for the arithmetic sequence with $a_1 = 10$, $d = 8$.

- a. 652
- b. 644
- c. 648 ✓
- d. 656

$$a_{12} = 10 + (11)(8) = 98. S_{12} = \frac{12(10 + 98)}{2} = 6(108) = 648.$$

26 Find S_{14} for the arithmetic sequence with $a_1 = 10$, $d = 6$.

a. 682

b. 686 ✓

c. 692

d. 690

$$a_{14} = 10 + (13)(6) = 88. S_{14} = \frac{14(10 + 88)}{2} = 7(98) = 686.$$

27 Find S_4 for the arithmetic sequence with $a_1 = 2$, $d = -7$.

a. -30

b. -41

c. -38

d. -34 ✓

$$a_4 = 2 + (3)(-7) = -19. S_4 = \frac{4(2 - 19)}{2} = 2(-17) = -34.$$

28 Find S_4 for the arithmetic sequence with $a_1 = -17$, $d = -5$.

a. -98 ✓

b. -102

c. -103

d. -94

$$a_4 = -17 + (3)(-5) = -32. S_4 = \frac{4(-17 - 32)}{2} = 2(-49) = -98.$$

29 Find S_{10} for the arithmetic sequence with $a_1 = 3$, $d = -8$.

a. -326

b. -330 ✓

c. -338

d. -334

$$a_{10} = 3 + (9)(-8) = -69. S_{10} = \frac{10(3 - 69)}{2} = 5(-66) = -330.$$

30 Find S_{12} for the arithmetic sequence with $a_1 = 14$, $d = -9$.

- a. -430
- b. -435
- c. -422
- d. -426 ✓

$$a_{12} = 14 + (11)(-9) = -85. S_{12} = \frac{12(14 - 85)}{2} = 6(-71) = -426.$$

31 Find S_8 for the arithmetic sequence with $a_1 = 1$, $d = -5$.

- a. -132 ✓
- b. -136
- c. -128
- d. -137

$$a_8 = 1 + (7)(-5) = -34. S_8 = \frac{8(1 - 34)}{2} = 4(-33) = -132.$$

32 Find S_{10} for the arithmetic sequence with $a_1 = -11$, $d = -6$.

- a. -376
- b. -386
- c. -380 ✓
- d. -384

$$a_{10} = -11 + (9)(-6) = -65. S_{10} = \frac{10(-11 - 65)}{2} = 5(-76) = -380.$$

33 Find S_{20} for the arithmetic sequence with $a_1 = -1$, $d = -4$.

a. -780 ✓

b. -774

c. -784

d. -776

$$a_{20} = -1 + (19)(-4) = -77. S_{20} = \frac{20(-1 - 77)}{2} = 10(-78) = -780.$$

34 Find S_{10} for the arithmetic sequence with $a_1 = 20$, $d = -7$.

a. -122

b. -115 ✓

c. -119

d. -111

$$a_{10} = 20 + (9)(-7) = -43. S_{10} = \frac{10(20 - 43)}{2} = 5(-23) = -115.$$

35 Find S_6 for the arithmetic sequence with $a_1 = -19$, $d = -1$.

a. -125

b. -133

c. -130

d. -129 ✓

$$a_6 = -19 + (5)(-1) = -24. S_6 = \frac{6(-19 - 24)}{2} = 3(-43) = -129.$$

36 Find S_8 for the arithmetic sequence with $a_1 = -15$, $d = -4$.

a. -228

b. -226

c. -236

d. -232 ✓

$$a_8 = -15 + (7)(-4) = -43. S_8 = \frac{8(-15 - 43)}{2} = 4(-58) = -232.$$

37 Find S_{12} for the arithmetic sequence with $a_1 = -13$, $d = -4$.

a. -424

b. -420 ✓

c. -414

d. -416

$$a_{12} = -13 + (11)(-4) = -57. S_{12} = \frac{12(-13 - 57)}{2} = 6(-70) = -420.$$

38 Find S_{16} for the arithmetic sequence with $a_1 = -12$, $d = 3$.

a. 168 ✓

b. 164

c. 171

d. 172

$$a_{16} = -12 + (15)(3) = 33. S_{16} = \frac{16(-12 + 33)}{2} = 8(21) = 168.$$

Multiple choice — hard

39 For the geometric sequence with $a_1 = -8$ and common ratio $r = 2$, find a_6 .

a. 256

b. -256 ✓

c. -258

d. -254

$$a_6 = a_1 r^5 = -8(2^5) = -8(32) = -256.$$

40 For the geometric sequence with $a_1 = -9$ and common ratio $r = -3$, find a_4 .

a. 243 ✓

b. 241

c. 245

d. -243

$$a_4 = a_1 r^3 = -9((-3)^3) = -9(-27) = 243.$$

41 For the geometric sequence with $a_1 = -2$ and common ratio $r = -3$, find a_6 .

a. -486

b. 488

c. 486 ✓

d. 484

$$a_6 = a_1 r^5 = -2((-3)^5) = -2(-243) = 486.$$

42 For the geometric sequence with $a_1 = -6$ and common ratio $r = -3$, find a_3 .

a. -54 ✓

b. 54

c. -56

d. -52

$$a_3 = a_1 r^2 = -6((-3)^2) = -6(9) = -54.$$

43 For the geometric sequence with $a_1 = 9$ and common ratio $r = 3$, find a_3 .

a. -81

b. 79

c. 81 ✓

d. 83

$$a_3 = a_1 r^2 = 9(3^2) = 9(9) = 81.$$

44 Find S_5 for the geometric sequence with $a_1 = -3$, $r = 3$.

a. -361

b. -363 ✓

c. -366

d. -365

$$S_5 = \frac{a_1(3^5 - 1)}{r - 1} = \frac{-3(242)}{2} = -363.$$

45 Find S_4 for the geometric sequence with $a_1 = -6$, $r = 3$.

a. -240 ✓

b. -238

c. -246

d. -242

$$S_4 = \frac{a_1(3^4 - 1)}{r - 1} = \frac{-6(80)}{2} = -240.$$

46 Find S_7 for the geometric sequence with $a_1 = -5$, $r = 3$.

a. -5467

b. -5465 ✓

c. -5470

d. -5463

$$S_7 = \frac{a_1(3^7 - 1)}{r - 1} = \frac{-5(2186)}{2} = -5465.$$

47 Find S_6 for the geometric sequence with $a_1 = 8$, $r = 3$.

a. 2920

b. 2912 ✓

c. 2914

d. 2910

$$S_6 = \frac{a_1(3^6 - 1)}{r - 1} = \frac{8(728)}{2} = 2912.$$

48 Find S_5 for the geometric sequence with $a_1 = -1$, $r = 2$.

a. -29

b. -33

c. -32

d. -31 ✓

$$S_5 = \frac{a_1(2^5 - 1)}{r - 1} = \frac{-1(31)}{1} = -31.$$

49 Find the sum of the infinite geometric series $216 - 36 + 6 \dots$.

a. $\frac{1296}{7}$ ✓

b. $\frac{1310}{7}$

c. $\frac{1282}{7}$

d. $-\frac{1296}{7}$

$$a = 216, r = -\frac{1}{6}, \text{ and } |r| < 1, \text{ so the sum is } \frac{216}{1 - \left(-\frac{1}{6}\right)} = \frac{1296}{7}.$$

50 Find the sum of the infinite geometric series $16 - 8 + 4 \dots$.

a. $\frac{38}{3}$

b. $\frac{32}{3}$ ✓

c. $\frac{26}{3}$

d. $-\frac{32}{3}$

$$a = 16, r = -\frac{1}{2}, \text{ and } |r| < 1, \text{ so the sum is } \frac{16}{1 - \left(-\frac{1}{2}\right)} = \frac{32}{3}.$$

51 Find the sum of the infinite geometric series $96 + 24 + 6\cdots$.

a. -128

b. 130

c. 126

d. 128 ✓

$a = 96$, $r = \frac{1}{4}$, and $|r| < 1$, so the sum is $\frac{96}{1 - \frac{1}{4}} = 128$.

52 Find the sum of the infinite geometric series $144 - 24 + 4\cdots$.

a. $\frac{878}{7}$

b. $\frac{850}{7}$

c. $\frac{864}{7}$ ✓

d. $-\frac{864}{7}$

$a = 144$, $r = -\frac{1}{6}$, and $|r| < 1$, so the sum is $\frac{144}{1 - \left(-\frac{1}{6}\right)} = \frac{864}{7}$.

53 Find the sum of the infinite geometric series $108 - 18 + 3\cdots$.

a. $-\frac{648}{7}$

b. $\frac{634}{7}$

c. $\frac{662}{7}$

d. $\frac{648}{7}$ ✓

$a = 108$, $r = -\frac{1}{6}$, and $|r| < 1$, so the sum is $\frac{108}{1 - \left(-\frac{1}{6}\right)} = \frac{648}{7}$.